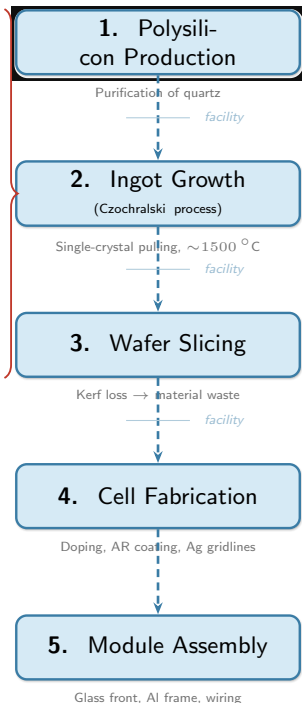


Monocrystalline c-Si

Disjointed, batch-based process
across specialized facilities



High-purity Si, Ag paste, kerf loss $\Rightarrow$ high variable cost per unit

Material Intensity

Energy per Unit

Furnaces at $\sim 1500^\circ\text{C}$ for days; energy cost scales linearly with volume

Active layer 100–300× thinner than Si wafer; lower deposition temp.

Low Material & Energy Input

Different cost optimization strategies

Cost/W driven by spot prices: polysilicon, silver, electricity

↑ **Variable-cost dominated**

= batch hand-offs between separate facilities

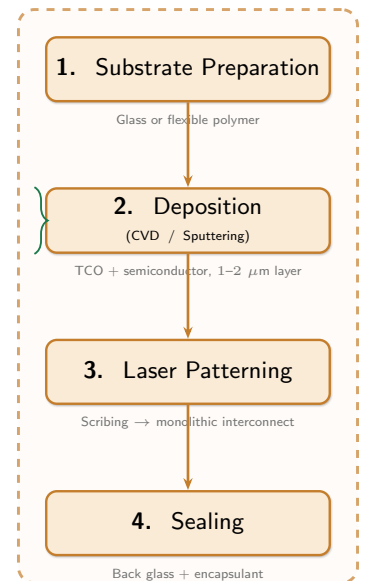
→ Solid arrows = continuous in-line flow

■ Variable-cost drivers

■ Fixed-cost drivers

Thin-Film (CdTe / CIGS)

Single Facility — Vertically Integrated
Continuous-flow process



Cost/W driven by utilization rate: must amortize CapEx over max MW

↑ **Fixed-cost dominated**

Capital Intensity (CapEx)

vacuum deposition chambers, continuous-flow robotics $\Rightarrow$ enormous upfront fixed cost